



NORSK DATA

PRODUCT INFORMATION

Product name

TCP/IP Basic Module/III

Product number

211327B

DESCRIPTION OF PRODUCT

To allow use of TCP/IP protocols by applications (FTP, TELNET) in a SINTRAN environment on a Ethernet local area network.

REASON(S) FOR NEW PRODUCT OR VERSION

Revision B05

- This product allows 124 simultaneous Telnet connections and 8 simultaneous FTP connections.
- SIBAS backend is now available over the TCP/IP Basic Module.
- NFS support may be selected when installing this product.

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1 PREREQUISITES

The following things are required in order to use this product.

1.1 HARDWARE PREREQUISITES

The following hardware must be present for this product to be used:

CPU type (any of the following):

- 5000

Floating format: *Any*

Other hardware:

Product name

Product number

Ethernet III controller

ND 110513

1.2 SOFTWARE PREREQUISITES

The following software must be installed on the system for this product to be used:

Operating system:

Name

Version/Revision

SINTRAN-III/VSX WM07
Standard Configuration A or B

L07

SINTRAN-VSX Patch file
Patch Level 4300

L04

ND-5200 microprogram
or

D01

ND-5400 microprogram
or

D01

ND-5500 microprogram
or

D01

ND-5700 microprogram
or

D01

ND-5800 microprogram

D01

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Other software:

Product nameProduct number

BACKUP-SYSTEM version >=I

210337

1.3 OTHER PREREQUISITES

The following are necessary for this product to be used.

Minimum Permanent Mass Storage:

User	Space (pages)	Number of files
SYSTEM	220	5
TCP-IP	234	13
UTILITY	188	1

	Minimum	Maximum
Number of segments (ND-100)	2	
Number of RT descriptions	2	

Space required on segment files: 72 pages

Requirements regarding MTAD resources per controller (one for each new connection used):

Max number of MTAD datafields: 124

Number of background programs: 124

Requirements regarding Nucleus resources for the first controller :

Message buffer space for system processes in pages: 16

Number of descriptors for all system processes: 32

Requirements regarding nucleus resources for each new controller:

Message buffer space for system processes in pages: 1

Number of descriptors for all system processes: 1

The max number of MTAD datafields (MTMAX) is changed with the SINTRAN-SERVICE-PROGRAM, the number of background programs and nucleus resources with the S3-CONFIG program.

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2 DEPENDENCIES

In addition to the above listed hardware and/or software prerequisites, the following dependencies exist between this and other products. The hardware and software listed here may, for example, be necessary to use the product's full functionality.

2.1 HARDWARE DEPENDENCIES

There are dependency relationships between this product and the following hardware:

None

2.2 SOFTWARE DEPENDENCIES

There are dependency relationships between this product and the following software:

Product nameProduct number

OWS Access Server for ND-5000

ND-211325

Nature of dependency

The installation of the OWS Access Server for ND-5000 depends on this product to be installed.

SIBAS/R Backend for ND-5000

ND-211404

Nature of dependency

This product must be installed if SIBAS Backend shall be used with TCP/IP communication.

2.3 OTHER DEPENDENCIES

In addition to the hardware and software dependency relationships listed above, you should be aware of the following dependencies:

None.

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3 DOCUMENTATION

Included (distributed free with product)

TitleNumber

OpenLAN Network Supervisor Guide

830107.03 EN

Optional (must be purchased)

TitleNumber

COSMOS TELNET/FTP User Guide

860284.01 EN

4 MAINTENANCE SOFTWARE

Included in the TCP/IP Basic Module/III is the TCP/IP-MONITOR, the Nucleus and Domino maintenance kits. They are intended for use by ND technical support. The TCP/IP-MONITOR is delivered as diskette no. 3 of the TCP/IP product, the others are independent products.

Product NameProduct Number

Nucleus Maintenance Kit version C

ND-211321

Domino Maintenance Kit version C

ND-211322

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5 INSTALLATION PROCEDURE

5.1 SELECTING INTERNET ADDRESS

- On request from the installation program you will have to assign an Internet address for each controller. If your net is isolated from other nets outside your company, you may select the address freely, i.e. using the default values given by the installation program. If your net connects to other nets, free selection of addresses may accidentally give the same address to two installations which will create severe problems. Selecting proper Internet addresses are further described in the OpenLAN Network Supervisor guide. However, the address for application of an official Internet address range is:

DDN Network Information Center
SRI International
Room EJ217
333 Ravenswood Avenue
Menlo Park, CA 94025

5.2 AUTOMATIC INSTALLATION

- Check that the RT-program PROMAN is running by giving the command:
@LIST-RT-DESCRIPTION PROMAN

If the program is not running, the first line will be:

PASSIVE

Start PROMAN by giving the command @START-SERVERS

If PROMAN starts without giving any error messages, the installation can continue.

If PROMAN fails to start, the command @START-SERVERS should be given in the LOAD-MODE:MODE or LOAD-SYSTEM:BATC as described in chapter 5.3, and the system restarted before installation is continued.

The installation will fail if PROMAN is passive.

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- Insert the floppy 211327B-XX-1D in the floppy-drive
@ENTER-DIRECTORY,211327B,F-D-1,(0/1)
- Log in as user SYSTEM
- Run the installation program:
@(211327:FLOPPY-USER)IN-TCP-BA
- The third floppy, 211327B-XX-03D contains Maintenance programs and is not required for normal use of the TCP/IP Basic Module.

5.3 COMPLETING THE INSTALLATION

When the installation program has been run :

- Update the AIP-HOSTS:SYMB file with the local internet address of the host name. See the documentation for more information.
- Update the mode file that is run after a cold start (usually called HENT-MODE:MODE) by including the following line:

@MODE (TCP-IP)TCP-BA-LOAD:MODE,,,

If you had a previous version of this product, remove the old line loading that version.

- Insert the line:

@START-SERVERS

if you have not already done it, in the LOAD-MODE:MODE or LOAD-SYSTEM:BATC file after starting the XMSG (and after the command @START-TADADM which usually lies in the XMSG startup file) and before starting the ND-5000. Be aware that some systems might have started the ND-5000 in the HENT-MODE:MODE job. In this case you should not start the ND-5000 in the HENT-MODE:MODE, but in the LOAD-MODE:MODE as mentioned above.

- To reboot the domino controller(s), run the mode job:
@MODE (TCP-IP)TCP-BA-REBO:M,,,
- To stop the domino controller(s), run the mode job:
@MODE (TCP-IP)TCP-BA-STOP:M,,,

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6 MODIFICATIONS

6.1 NEW FUNCTIONS/COMMANDS

- Possibility of running TCP/IP in several Ethernet III controllers simultaneously.
- An error message will be issued when another host has the same Internet address.
- This product will now support startup and error reporting to a NCS network server. The Internet address of the NCS server must be put into the AIP-HOSTS:SYMB file with the name AUDITSERVER to achieve this. The product will also answer alive check polling from the NCS server.

6.2 CHANGED FUNCTIONS/COMMANDS

None.

6.3 REMOVED FUNCTIONS/COMMANDS

None.

6.4 OTHER MODIFICATIONS

- The installation program will ask you whether or not you require NFS support. You should answer NO to this unless you are installing NFS at the same time, or else some TCP/IP resources will be wasted.
- The installation program will prompt for the number of Telnet server connections actually bought. You MUST give the correct number, as found in the accompanying ND Licence Agreement document.
- If you are re-installing this product you have to do a warm start afterwards to make the re-installed product function properly with respect to any changes to its configuration. For example, in terms of number of Telnet server connections.

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7 ERRORS CORRECTED

- The Telnet Server will now close a connection if it times out.
- A Telnet connection doing XOFF will not cause the connection to be closed.
- A Telnet connection will run more smoothly due to enhancements regarding queue handling and event handling in the TCP and the Telnet server.
- On power fail the TCP Control server in ND-100 will not close down the controller.
- The stability of the TCP/IP is improved: it will not go down due to long Ethernet packets on the net and it will not accidentally close its own port due to returned messages from applications.
- Dead connections will now be removed by a timer in the TCP.
- Earlier versions sometimes closed connection after 10 minutes without traffic. This was caused by an error in the No Activity Timer. This has been corrected in this version.
- In earlier versions, the FTP server was not capable of handling files larger than 16 MB. The maximum file size is now increased to 128 MB.
- When doing serial file transfer using the OWS Terminal Line Server over a Telnet connection, older versions would sometimes fail when transferring files to the server. This has now been corrected.
- When using SIBAS Backend over TCP/IP, Sibas connections would sometimes be closed down with an error message from the routine HandleSibdata. The error occurred now and then when using the call STOWD (Storing of wide columns).

8 ERRORS KNOWN BUT NOT CORRECTED

None.